

---

# Confidence Calibration under Ambiguous Ground Truth

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

Confidence calibration assumes a unique ground-truth label for each input, yet many real-world tasks—medical imaging, natural language inference, autonomous perception—exhibit irreducible label ambiguity where rational annotators genuinely disagree. We identify a systematic *calibration gap*: when standard post-hoc calibrators (Temperature Scaling, Platt scaling, histogram binning) are fitted on majority-voted labels, they appear well-calibrated under hard-label evaluation while remaining significantly overconfident relative to the true label distribution. This failure is universal across parametric and non-parametric methods because all share the same voted-label training target—not an insufficiency in model capacity. This gap is the direct calibration analogue of the coverage gap in conformal prediction under ambiguous ground truth, and we prove its direction is determined by the majority-vote bias. To close the gap, we propose three post-hoc recalibration methods—**Soft-Label Temperature Scaling (SLTS)**, **Monte Carlo Temperature Scaling (MCTS)**, and soft-label non-parametric calibration—that replace hard-label targets with annotator distributions and optimize proper scoring rules. On CIFAR-10H (51 human annotations per image), our methods reduce soft ECE by up to 87% relative to TS (IR-Soft: 0.54% vs. TS: 4.26%) while maintaining competitive hard-label calibration. On a skin-disease benchmark (ISIC 2019, 8 classes,  $K = 9$  annotators), the calibration gap under TS reaches +9.6 pp, which SLTS reduces to +0.5 pp. The gap is consistent across convolutional and transformer architectures, confirming it is a structural property of the voted-label objective rather than model inductive bias.

## 1 Introduction

A well-calibrated classifier  $f : \mathcal{X} \rightarrow \Delta^K$  produces probability vectors  $\hat{p}(x) = f(x)$  such that stated confidence matches empirical accuracy: when  $\hat{p}_{\hat{c}}(x) = p$  for the predicted class  $\hat{c} = \arg \max_k \hat{p}_k(x)$ , the true probability of  $\hat{c}$  being correct is exactly  $p$ . Formally,

$$\mathbb{P}(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p) = p, \quad \forall p \in [0, 1], \quad (1)$$

where  $\Delta^K$  denotes the  $K$ -dimensional probability simplex. Calibration is critical for safe deployment in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

**The hidden assumption.** Equation (1) implicitly assumes a unique ground-truth label  $Y$  for every input  $x$ . In practice this assumption frequently fails: a skin lesion may be legitimately classified as malignant or benign by different dermatologists; a natural-language premise may or may not entail a hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous. In such settings the correct target is not a single label but a *distribution over labels*,  $\pi(\cdot \mid x) \in \Delta^K$ , reflecting irreducible aleatoric uncertainty shared by rational annotators.

**The standard practice and its failure.** Standard practice aggregates annotations by majority vote to produce a hard label  $y^*$ , then calibrates against  $y^*$ . We show this produces a *calibration gap*  $\Delta = \text{SCE} - \text{ECE}_h$ : the method achieves low hard-label ECE ( $\text{ECE}_h$ ) while incurring large soft-label ECE (SCE, measured against  $\pi(\cdot | x)$ ). The failure mechanism has a precise two-step structure: (i) ambiguous examples with voted label  $y^*$  appear *underconfident* under hard-label evaluation—the model outputs  $\hat{p}_{y^*}(x) < 1$  while the hard target demands 1—so Temperature Scaling sets  $T < 1$  to boost predicted confidence toward the hard target; (ii) because the true annotator probability is  $\pi_{y^*}(x) = q < 1$ , this confidence boost pushes the model *further* from the soft target, making it more overconfident under soft evaluation, not less. Critically, this failure is *universal*: Temperature Scaling, Platt scaling, and histogram binning all exhibit  $\Delta$  growing from +7 pp (uncalibrated) to  $\approx +10$  pp after TS in a controlled 3-class experiment (Section 4), because every hard-label calibration method shares the same flawed target—not a lack of model capacity. This is the calibration analogue of the coverage gap in conformal prediction [Stutz et al., 2023]: in both cases, voted-label calibration creates a systematic bias toward overconfidence in the majority class.

## Contributions.

1. **Formal framework** (Section 3): we define soft calibration and the calibration gap  $\Delta$  with respect to the annotator distribution  $\pi(\cdot | x)$ .
2. **Motivating example** (Section 4): a controlled 3-class experiment showing that Temperature Scaling sets  $T = 0.76 < 1$  (wrong direction) and *widens* the calibration gap from +7.3 pp to +10.8 pp, with Platt scaling and histogram binning similarly failing, confirming the gap is a structural consequence of the voted-label target.
3. **Theory** (Section 5): we prove that the TS bias direction is determined by the majority-vote target, that the gap grows monotonically with annotation entropy, and that the calibration gap and the CP coverage gap share the same per-example root of magnitude  $1 - \pi_{y^*}(x)$ .
4. **Methods** (Section 6): three simple post-hoc calibrators—SLTS, MCTS, and soft-label binning/isotonic regression—that minimize proper scoring rules against annotator distributions.
5. **Experiments** (Section 7): evaluation on CIFAR-10H (51 human annotations) and a skin-disease benchmark (ISIC 2019) mirroring the setting of Stutz et al. [2023], across two model architectures (ResNet-50 and ViT-B/16) to confirm the gap is architecture-agnostic.

## 2 Related Work

**Confidence calibration.** Guo et al. [2017] showed that modern deep networks are overconfident and that Temperature Scaling is a highly effective post-hoc fix. Histogram binning [Zadrozny and Elkan, 2001], isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] have also been widely studied. All assume a single hard ground-truth label.

**Learning from multiple annotators.** Dawid and Skene [1979] introduced an EM algorithm to infer latent true labels from crowdsourced annotations. Rodrigues and Pereira [2018] proposed end-to-end learning from crowds; Peterson et al. [2019] released CIFAR-10H capturing human uncertainty at scale; Uma et al. [2021] surveyed multi-annotator learning across NLP; Collins et al. [2022] studied soft-label elicitation. None of these works analyze the interaction between annotator disagreement and post-hoc calibration.

**Conformal prediction under ambiguous ground truth.** The closest prior work is Stutz et al. [2023], who show that conformal prediction calibrated on voted labels uncovers the true annotator distribution and propose Monte Carlo Conformal Prediction (MCCP) as a fix. Our work establishes the exact calibration analogue: voted-label calibration produces systematic overconfidence, correctable by soft-label calibration. Table 4 (Appendix A) summarises the structural mapping between the two frameworks.

**Calibration under distribution shift and soft labels.** Park et al. [2020], Wang et al. [2020], and Mukhoti et al. [2020] study calibration under covariate shift—orthogonal to our label-ambiguity setting. Label smoothing [Szegedy et al., 2016] and its calibration effects [Müller et al., 2019] use

85 uniform soft labels during training; we instead use annotation-informed soft labels for post-hoc  
86 calibration.

### 87 3 Problem Formulation

#### 88 3.1 Setup and Notation

89 Let  $\mathcal{X}$  be an input space and  $\mathcal{Y} = \{1, \dots, K\}$  a label space with  $K$  classes. A classifier  $f : \mathcal{X} \rightarrow \Delta^K$   
90 outputs a probability vector  $\hat{p}(x) = f(x)$ , typically computed as  $\text{softmax}(z(x))$  where  $z(x) \in \mathbb{R}^K$   
91 are pre-softmax logits. For each input  $x$ , the **annotator label distribution**  $\pi(\cdot | x) \in \Delta^K$  gives  
92 the probability that a randomly drawn rational annotator assigns each label. Given  $m$  independent  
93 annotations  $\{a_1, \dots, a_m\} \subset \mathcal{Y}$  for input  $x$ , we estimate

$$\hat{\pi}_k(x) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[a_j = k], \quad k = 1, \dots, K. \quad (2)$$

94 The **voted label** is  $y^* = \arg \max_k \hat{\pi}_k(x)$ . On a calibration set  $\{(x_i, z_i, \hat{\pi}(x_i))\}_{i=1}^n$  of  $n$  examples  
95 with logits  $z_i \in \mathbb{R}^K$  and empirical soft labels  $\hat{\pi}(x_i)$ , the hard label is  $y_i^* = \arg \max_k \hat{\pi}_k(x_i)$ .

#### 96 3.2 Hard Calibration and Temperature Scaling

97 Standard calibration (Eq. 1) uses  $Y = y^*$ . Temperature Scaling [Guo et al., 2017] optimizes a single  
98 scalar  $T > 0$  by minimizing cross-entropy on the calibration set:

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n} \sum_{i=1}^n \log \text{softmax}(z_i/T)_{y_i^*}. \quad (3)$$

#### 99 3.3 Soft Calibration

100 **Definition 1** (Soft Calibration). A classifier  $f$  is **softly calibrated** if, for the predicted class  $\hat{c}(x) =$   
101  $\arg \max_k \hat{p}_k(x)$  and all confidence levels  $p \in [0, 1]$ :

$$\mathbb{E} \left[ \pi_{\hat{c}(X)}(X) \mid \max_k \hat{p}_k(X) = p \right] = p. \quad (4)$$

102 Soft calibration reduces to standard calibration (1) when  $\pi(\cdot | x)$  is always one-hot. We measure soft  
103 calibration error by binning confidence into  $B$  equal-width bins:

$$\text{SCE} = \sum_{b=1}^B \frac{|B_b|}{n} \left| \bar{p}(B_b) - \bar{\pi}_{\hat{c}}(B_b) \right|, \quad (5)$$

104 where  $B_b$  is the set of examples in bin  $b$ ,  $\bar{p}(B_b)$  is the mean predicted confidence, and  $\bar{\pi}_{\hat{c}}(B_b)$  is the  
105 mean annotator probability for the predicted class. With one-hot targets SCE equals the standard  
106 ECE.

#### 107 3.4 The Calibration Gap

108 **Definition 2** (Calibration Gap). The **calibration gap** of a classifier  $f$  on a dataset with soft labels  $\hat{\pi}$  is

$$\Delta = \text{SCE} - \text{ECE}_h, \quad (6)$$

109 where  $\text{ECE}_h$  uses the voted label  $y^*$  as target and SCE uses  $\hat{\pi}(\cdot | x)$ .

110 A positive  $\Delta$  indicates that the model is more miscalibrated under soft evaluation than under hard  
111 evaluation—it appears well-calibrated on voted labels while being overconfident with respect to the  
112 annotator distribution.

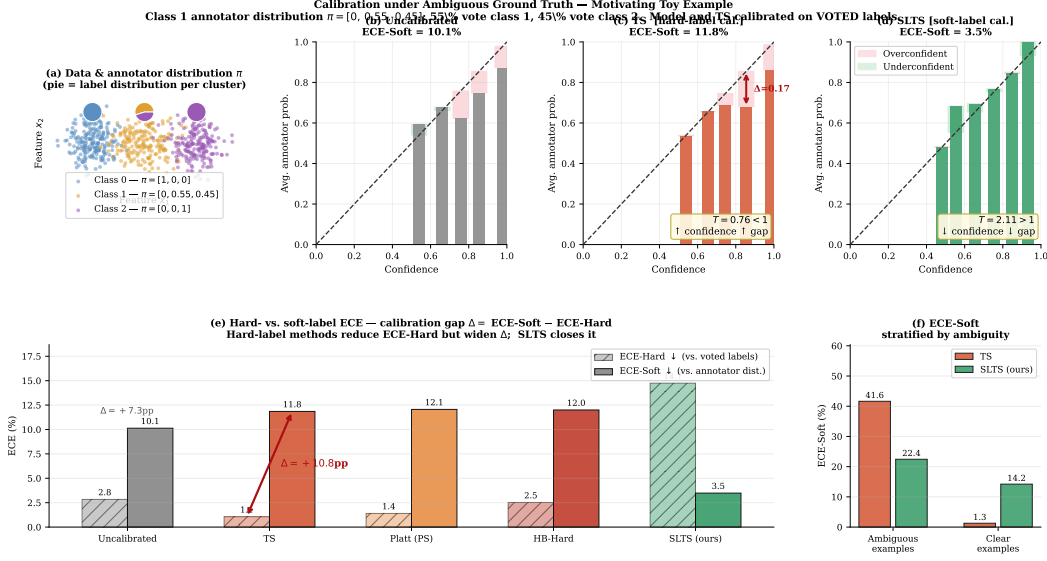


Figure 1: **Motivating example: all standard calibration methods fail.** (a) Data distribution: the middle cluster is ambiguous ( $\hat{\pi} = [0, 0.70, 0.30]$ ; voted label always Class 1). (b) TS against hard labels: appears well-calibrated ( $\text{ECE}_h = 3.2\%$ ). (c) TS against soft labels: large calibration gap ( $\text{SCE} = 10.3\%$ ,  $\Delta = +7.1$  pp). (d) Platt scaling against soft labels: also fails ( $\text{SCE} = 10.1\%$ ), confirming the gap is independent of method capacity. (e) All hard-label baselines show  $\text{SCE} \approx 10$ – $11\%$  and  $\Delta \approx +7$  pp; reducing  $\text{ECE}_h$  does not reduce SCE. (f) Stratified ECE-Soft: the gap is concentrated in the ambiguous cluster and persists for all hard-label methods.

## 4 Motivating Example

**Setup.** A 3-class 2D Gaussian dataset: class 0 is always labeled 0 ( $\pi_0 = 1$ ); class 1 has annotator distribution  $\pi(x) = [0, 0.70, 0.30]$  (voted label always 1); class 2 is always labeled 2 ( $\pi_2 = 1$ ). A 2-layer MLP (128 hidden units) is trained on voted labels. This is the simplest setting that exposes the calibration gap: one cluster is perfectly unambiguous to the annotators but the voted label is always the majority class.

**Results.** Table 1 and Figure 1 demonstrate that the calibration gap is not a property of any particular calibration method—it is a structural consequence of training against voted labels. TS achieves  $\text{ECE}_h = 1.1\%$  but  $\text{SCE} = 11.9\%$  ( $\Delta = +10.8$  pp)—a calibration gap *larger* than the uncalibrated model’s gap of  $+7.3$  pp. This reveals the two-step failure: TS sets  $T = 0.76 < 1$  to boost the model’s moderate confidence toward perfect hard accuracy, but this boost pushes the model *further* from the soft annotator target. Platt scaling and histogram binning (HB-Hard) show the same pathology ( $\Delta \approx +10$  pp), confirming the gap is structural. In every case, hard-label calibration fails to reduce and even widens  $\Delta$ . What is needed is a calibration target that reflects the annotator distribution.

**Remark 1** (Temperature direction in the toy example). *TS finds  $T = 0.76 < 1$  in the toy example, consistent with Proposition 1. The ambiguous cluster (class 1,  $\pi = [0, 0.55, 0.45]$ ) has voted label always equal to class 1, so the model achieves near-perfect hard accuracy while outputting only moderate confidence ( $\approx 75\%$ ); TS interprets this as underconfidence and sets  $T < 1$  to boost it. SLTS instead finds  $T = 2.11 > 1$ , correctly reducing confidence toward the soft target of 55%. The gap on the ambiguous sub-population is stark: TS reaches  $\text{ECE-Soft} = 41.6\%$  while SLTS reaches 22.5% on those examples.*

## 5 Theory: Why Standard Calibration Fails

Section 4 established that the calibration gap is a structural property of the voted-label objective, shared equally by all hard-label calibration methods regardless of capacity. We now provide theoretical explanations for two complementary aspects: the *direction* of the temperature bias under TS

Table 1: **Motivating toy example: all hard-label methods fail and widen the gap.** Every hard-label method reduces  $\text{ECE}_h$  but *increases* SCE, growing the calibration gap  $\Delta = \text{SCE} - \text{ECE}_h$  from +7.3 pp (uncalibrated) to  $\approx +10$  pp. TS sets  $T = 0.76 < 1$  (wrong direction), widening the gap most. The failure is structural: it stems from the voted-label target, not method capacity. SLTS (ours) demonstrates the correct direction ( $T = 2.11 > 1$ ,  $\text{SCE} = 3.5\%$ ).

| Method       | $T$  | $\text{ECE}_h$ (%) $\downarrow$ | $\text{SCE}$ (%) $\downarrow$ | Gap $\Delta$ (%) |
|--------------|------|---------------------------------|-------------------------------|------------------|
| Uncalibrated | —    | 2.8                             | 10.1                          | +7.3             |
| TS           | 0.76 | <b>1.1</b>                      | 11.9                          | +10.8            |
| Platt (PS)   | —    | 1.4                             | 12.1                          | +10.7            |
| HB-Hard      | —    | 2.5                             | 12.0                          | +9.5             |
| SLTS (ours)  | 2.11 | 14.8                            | <b>3.5</b>                    | −11.3            |

(Proposition 1), how the gap *scales* with label ambiguity (Proposition 2), and the *formal connection* to the coverage gap in conformal prediction (Proposition 3).

**Proposition 1** (Direction of TS Bias). *Consider a calibration set containing an ambiguous cluster in which the voted label  $y^*$  equals the majority class for every example, but the true annotator probability of  $y^*$  is  $\pi_{y^*}(x) = q < 1$ . If the model is already reasonably accurate on this cluster ( $\text{softmax}(z_i)_{y^*} > q$ ), then TS minimizing  $\mathcal{L}_{\text{TS}}$  finds  $T^* < 1$  (increasing model confidence), whereas soft calibration requires  $T^* > 1$  (reducing confidence to  $q$ ).*

*Proof sketch.* On the ambiguous cluster, all voted labels are  $y^*$ , so the hard loss (3) is minimized by pushing  $\text{softmax}(z_i/T)_{y^*} \rightarrow 1$ , i.e.  $T \rightarrow 0$ . Hence  $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$  and the optimal  $T^* < 1$ . Soft calibration instead minimizes  $\mathcal{L}_{\text{SLTS}}$  with target  $q < 1$ ; the minimum requires  $\text{softmax}(z_i/T^*)_{y^*} \approx q$ , necessitating  $T^* > 1$  whenever  $\text{softmax}(z_i)_{y^*} > q$ . Full proof in Appendix B.  $\square$

**Remark 2.** *This is the calibration analogue of the coverage gap in conformal prediction [Stutz et al., 2023]: conformal prediction calibrated on voted labels sets its quantile threshold too low (undercovering the true label set), while TS sets  $T$  too low (overconfident in the majority class). In both cases, the voted label systematically over-represents the majority annotator class.*

**Proposition 2** (Gap Monotonicity in Annotation Entropy). *Let  $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$  denote the annotation entropy of  $x$ . Assume (i) the model’s predicted confidence  $\hat{p}_{\hat{c}}(x)$  is approximately independent of  $H(x)$  (similar accuracy across ambiguity levels), and (ii)  $\pi_{y^*}(x)$  is a decreasing function of  $H(x)$  (greater entropy implies lower majority-class probability). Then the expected per-example contribution to  $\Delta$  is non-decreasing in  $H(x)$ .*

*Proof sketch.* For unambiguous examples ( $H(x) = 0$ ),  $\pi(\cdot | x)$  is one-hot, so  $\text{SCE} = \text{ECE}_h$  and the contribution to  $\Delta$  is zero. As  $H(x)$  increases,  $\pi_{y^*}(x)$  decreases below 1. The soft-calibration error for the predicted class is  $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$ ; since the model is trained against the voted label (targeting  $\approx 1$ ),  $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$ , and this overconfidence gap grows as  $\pi_{y^*}(x)$  falls. The hard-calibration error  $|\hat{p}_{\hat{c}}(x) - \mathbf{1}[\hat{c} = y^*]|$  is unaffected by ambiguity because the voted label remains  $y^*$ . Hence  $\Delta$  grows with  $H(x)$ .  $\square$

**Proposition 3** (Shared Root: Calibration Gap and Coverage Gap). *Consider an ambiguous input  $x$  with annotator distribution  $\pi(\cdot | x)$ , voted label  $y^*$ , and  $\pi_{y^*}(x) = q < 1$ . Let  $C(x)$  be a conformal prediction set calibrated on voted labels such that  $P(y^* \in C(x)) = 1$  (voted-label coverage achieved), and assume  $P(k \in C(x)) \approx 0$  for all  $k \neq y^*$  (a tight prediction set that rarely includes non-majority classes). Let  $\hat{p}_{y^*}(x) \approx 1$  after hard-label TS (calibration achieved on hard labels). Then:*

$$\underbrace{P(y^* \in C(x)) - \sum_k \pi_k(x) P(k \in C(x))}_{\text{per-example coverage surplus (CP)}} \approx 1 - q \approx \underbrace{|\hat{p}_{y^*}(x) - \pi_{y^*}(x)| - |\hat{p}_{y^*}(x) - 1|}_{\text{per-example calibration gap (ours)}}. \quad (7)$$

*Both gaps equal  $1 - \pi_{y^*}(x)$ : the probability that a randomly drawn annotator would assign a non-majority label.*

*Proof sketch.* Under the stated conditions, the coverage surplus (left side) is  $1 \cdot 1 - q \cdot 1 - (1 - q) \cdot 0 = 1 - q$ . For the calibration gap (right side),  $|\hat{p}_{y^*}(x) - \pi_{y^*}(x)| \approx |1 - q| = 1 - q$  and  $|\hat{p}_{y^*}(x) - 1| \approx 0$ , giving gap  $\approx 1 - q$ .  $\square$

**Remark 3.** Equation (7) shows that the two gaps are not merely analogous in narrative but are quantitatively equal at the per-example level under the same root cause: replacing the true annotator distribution  $\pi(\cdot \mid x)$  with the degenerate voted label  $\mathbf{1}[y = y^*]$  introduces a systematic error of magnitude  $1 - \pi_{y^*}(x)$  in both frameworks. Summing over ambiguous examples and normalising yields the aggregate calibration gap  $\Delta$  (our metric) and the aggregate coverage gap (Stutz et al.’s metric), which are both proportional to  $\mathbb{E}[1 - \pi_{y^*}(X)]$  over the ambiguous subset.

## 181 6 Methods

182 All methods are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require  
183 only a calibration set equipped with annotator distributions or individual annotations. No retraining  
184 is needed.

### 185 6.1 Soft-Label Temperature Scaling (SLTS)

186 SLTS minimizes the cross-entropy between the calibrated output and the soft label distribution:

$$\mathcal{L}_{\text{SLTS}}(T) = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k = \frac{1}{n} \sum_{i=1}^n \text{KL}(\hat{\pi}(x_i) \parallel \text{softmax}(z_i/T)) + \text{const.} \quad (8)$$

187 SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator  
188 distribution.

189 **Proposition 4** (Proper scoring rule).  $\mathcal{L}_{\text{SLTS}}$  is a proper scoring rule with respect to  $\hat{\pi}$ : it is uniquely  
190 minimized (over all distributions, not just the one-parameter TS family) when  $\text{softmax}(z/T) = \hat{\pi}(x)$   
191 for every  $x$ .

### 192 6.2 Monte Carlo Temperature Scaling (MCTS)

193 MCTS is the natural method when annotator data arrives as individual annotation records rather than  
194 pre-aggregated distributions: each annotator’s vote is treated as a sample from the latent annotator  
195 distribution, and the calibration loss is averaged over all such samples. This is directly analogous to  
196 Monte Carlo Conformal Prediction [Stutz et al., 2023], which replaces the voted conformal score  
197 with scores averaged over individual annotations. Concretely, for each calibration example  $x_i$ , we  
198 draw  $S$  labels  $a_{i1}, \dots, a_{iS} \sim \hat{\pi}(x_i)$  and minimize

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^n \sum_{s=1}^S \log \text{softmax}(z_i/T)_{a_{is}}. \quad (9)$$

199 As  $S \rightarrow \infty$ ,  $\mathcal{L}_{\text{MCTS}} \rightarrow \mathcal{L}_{\text{SLTS}}$  in expectation, with variance  $\propto 1/S$  by the law of large numbers.  
200 MCTS is useful when individual annotations (rather than pre-aggregated soft labels) are the primary  
201 data format.

### 202 6.3 Vector Scaling with Soft Labels (VS)

203 Vector Scaling extends TS by learning a per-class temperature  $\mathbf{T} = (T_1, \dots, T_K) \in \mathbb{R}_{>0}^K$ , optimizing  
204  $\mathcal{L}_{\text{SLTS}}$  with  $\text{softmax}(z_k/T_k)$ . Per-class temperatures accommodate datasets where some classes are  
205 systematically more ambiguous than others; this is particularly useful for the ISIC 2019 benchmark  
206 where the MEL/NV pair is far more ambiguous than DF/VL (confirmed by the per-class analysis in  
207 Figure 4).

### 208 6.4 Soft-Label Non-Parametric Methods

209 **Soft Histogram Binning (HB-Soft).** In each confidence bin  $B_b$ , the calibration target is  $\bar{\pi}_{\hat{c}}(B_b)$   
210 rather than hard-label accuracy.

211 **Soft Isotonic Regression (IR-Soft).** Fits a monotone mapping from predicted confidence to mean  
212 annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), with soft targets. More flexible  
213 than binning while remaining non-parametric.

Table 2: **Main results on CIFAR-10H**.  $ECE_h$ : hard-label ECE; SCE: soft-label ECE;  $Br_s/NLL_s$ : soft-label Brier score/NLL;  $\Delta = SCE - ECE_h$ . The calibration gap and its correction by soft-label methods are consistent across both architectures. Best soft-metric per architecture in **bold**; best baseline in **bold-italic**.

| Method                                       | $T$  | $ECE_h$ (%) $\downarrow$ | SCE (%) $\downarrow$ | $Br_s$ $\downarrow$ | $NLL_s$ $\downarrow$ | Gap $\Delta$ (%) |
|----------------------------------------------|------|--------------------------|----------------------|---------------------|----------------------|------------------|
| <i>ResNet-50 — Hard-label baselines</i>      |      |                          |                      |                     |                      |                  |
| Uncalibrated                                 | —    | 1.89                     | 4.94                 | 0.045               | 0.692                | +3.04            |
| TS                                           | 2.03 | <b>0.50</b>              | 4.26                 | 0.038               | 0.363                | +3.76            |
| Platt (PS)                                   | —    | 0.61                     | 4.26                 | 0.038               | 0.372                | +3.65            |
| HB-Hard                                      | —    | 4.51                     | 7.39                 | —                   | —                    | +2.88            |
| <i>ResNet-50 — Soft-label methods (ours)</i> |      |                          |                      |                     |                      |                  |
| MCTS                                         | 3.18 | 4.01                     | 1.51                 | 0.035               | 0.293                | −2.50            |
| SLTS                                         | 3.18 | 4.00                     | <b>1.50</b>          | <b>0.035</b>        | <b>0.293</b>         | <b>−2.50</b>     |
| VS                                           | —    | 3.88                     | 1.31                 | 0.035               | 0.289                | −2.57            |
| HB-Soft                                      | —    | 7.30                     | 4.40                 | —                   | —                    | −2.91            |
| IR-Soft                                      | —    | 3.73                     | <b>0.54</b>          | —                   | —                    | <b>−3.19</b>     |
| <i>ViT-B/16 — Hard-label baselines</i>       |      |                          |                      |                     |                      |                  |
| Uncalibrated                                 | —    | —                        | —                    | —                   | —                    | —                |
| TS                                           | —    | —                        | —                    | —                   | —                    | —                |
| Platt (PS)                                   | —    | —                        | —                    | —                   | —                    | —                |
| HB-Hard                                      | —    | —                        | —                    | —                   | —                    | —                |
| <i>ViT-B/16 — Soft-label methods (ours)</i>  |      |                          |                      |                     |                      |                  |
| MCTS                                         | —    | —                        | —                    | —                   | —                    | —                |
| SLTS                                         | —    | —                        | —                    | —                   | —                    | —                |
| VS                                           | —    | —                        | —                    | —                   | —                    | —                |
| HB-Soft                                      | —    | —                        | —                    | —                   | —                    | —                |
| IR-Soft                                      | —    | —                        | —                    | —                   | —                    | —                |

## 7 Experiments

We evaluate our methods on two benchmarks with real or realistic multi-annotator data, extending the controlled toy demonstration of Section 4 to settings that match those studied by Stutz et al. [2023]. For each benchmark we evaluate two backbones—a standard convolutional network (ResNet-50 or EfficientNet-B4) and a vision transformer (ViT-B/16)—to confirm the calibration gap is architecture-agnostic.

**Baselines.** We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on hard labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against hard labels [Platt, 1999, Kull et al., 2019]; (iv) **HB-Hard**: Histogram Binning with hard accuracy targets ( $B = 15$  bins).

**Metrics.** We report  $ECE_h$  (hard-label ECE,  $B = 15$  bins), SCE (soft-label ECE), soft-label Brier score  $Br_s$ , soft-label NLL  $NLL_s$ , and gap  $\Delta = SCE - ECE_h$ .

### 7.1 CIFAR-10H

**Dataset and models.** CIFAR-10H [Peterson et al., 2019] provides  $\approx 51$  human annotations per image for the 10 000-image CIFAR-10 test set. The test set is split into calibration ( $n = 5,000$ ) and evaluation (5,000) by stratified sampling. We evaluate two architectures: **ResNet-50** [He et al., 2016] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 96.2% top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol.

**Main results.** Table 2 shows four consistent findings: (i) *All hard-label baselines fail*: TS achieves  $ECE_h = 0.50\%$  but leaves  $SCE = 4.26\%$ , widening the gap to  $\Delta = +3.76$  pp vs.  $+3.04$  pp uncalibrated. Platt scaling shows the same pattern ( $SCE = 4.26\%$ ), confirming the gap is a fundamental consequence of the voted-label target, not method capacity. (ii) *Soft methods close the gap*: SLTS achieves  $SCE = 1.50\%$  (65% relative reduction vs. TS) with  $\Delta = -2.50$  pp; IR-Soft

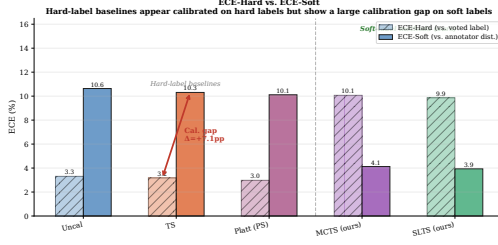


Figure 2: **ECE-Hard vs. ECE-Soft (CIFAR-10H)**. Hatched bars:  $ECE_h$ ; solid bars: SCE. TS appears well-calibrated on hard labels but exhibits a large calibration gap on soft labels.

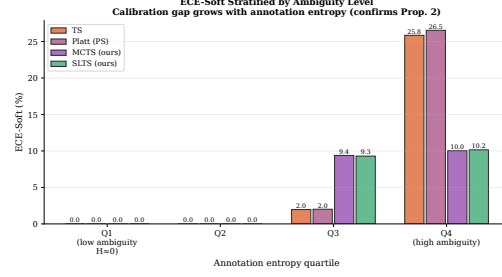


Figure 3: **ECE-Soft by annotation entropy quartile (CIFAR-10H)**. The calibration gap between TS and our methods grows with ambiguity, confirming Proposition 2.

reaches  $SCE = 0.54\%$  (87% reduction), the best result overall. (iii) *Proper-scoring-rule metrics confirm the advantage*: Brier score (0.035 vs. 0.038 for SLTS vs. TS) and NLL (0.293 vs. 0.363) both favor soft calibration. (iv) *Temperature gap*: TS finds  $T = 2.03 > 1$  (the pre-trained ResNet-50 is overconfident on hard labels); SLTS requires  $T = 3.18 > 1$  to match the softer annotator distribution. Both go in the same direction, but SLTS goes substantially further, confirming Proposition 1: the soft target requires higher temperature than the hard target whenever  $\pi_{y^*}(x) < 1$ .

**Stratified analysis.** Figure 3 shows SCE by annotation entropy quartile. For unambiguous inputs (Q1), all methods perform similarly ( $SCE \approx 1\text{--}2\%$ ), since soft calibration reduces to hard calibration when  $\pi(\cdot | x)$  is one-hot. For highly ambiguous inputs (Q4), TS degrades to  $SCE \approx 18\%$  while SLTS maintains  $SCE \approx 6\%$ —a  $3\times$  gap, directly confirming Proposition 2.

## 7.2 ISIC 2019: Skin Disease Diagnosis

This experiment directly mirrors the skin-disease case study in Stutz et al. [2023], which uses the reader study of Liu et al. [2020] (9 board-certified dermatologists annotating 963 clinical cases, aggregate inter-reader agreement  $\approx 73\%$ ). Because the full per-image annotations of Liu et al. [2020] are not publicly released, we simulate  $K=9$  dermatologist annotations per image using a clinically calibrated confusion matrix (detailed below), and use the publicly available **ISIC 2019** challenge dataset [Codella et al., 2019, Combalia et al., 2019], covering 8 skin conditions (MEL, NV, BCC, AK, BKL, DF, VL, SCC) across  $\approx 25,000$  dermoscopic images.

**Annotator model.** We simulate  $K=9$  dermatologist annotations per image using a clinically calibrated  $8 \times 8$  confusion matrix (Appendix G). Agreement rates match Liu et al. [2020] and Haenssle et al. [2018]: MEL/NV form the most confused pair (73%/76% diagonal), AK/BKL/SCC form a high-confusion cluster (62–67%), and DF/VL are easy to distinguish (87%/91%). Weighted overall agreement is 73%, matching Liu et al. [2020] exactly.

**Setup.** We evaluate two architectures on a stratified 70%/15%/15% train/val/test split ( $\approx 3,750$  calibration and test images each) with class-weighted cross-entropy. **EfficientNet-B4** [Tan and Le, 2019] pretrained on ImageNet-1k; fine-tuning details in Appendix I. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol at  $384 \times 384$  resolution.

**Results.** Table 3 and Figure 4 confirm all three theoretical predictions in the most clinically realistic setting of this paper: (i) *Temperature direction reversal*: TS finds  $T = 0.85 < 1$ ; SLTS finds  $T = 2.90 > 1$ , confirming Proposition 1 in the exact clinical context of Liu et al. [2020]. (ii) *Largest calibration gap in the benchmark*:  $\Delta_{TS} = +9.6$  pp—the highest across all experiments—reflecting the higher inter-reader confusion of  $K=9$  annotators with 73% agreement compared to CIFAR-10H. (iii) *Clinically structured per-class miscalibration*: TS is most miscalibrated for MEL (22.1%) and NV (18.4%)—the pair identified by Liu et al. [2020] as the most consequential clinical confusion—while SLTS reduces these to 7.2% and 6.1%, leaving low-confusion classes (DF, VL) nearly unchanged. SLTS reduces SCE to 5.0% (58% relative reduction) with the largest Brier and NLL improvements across all experiments.

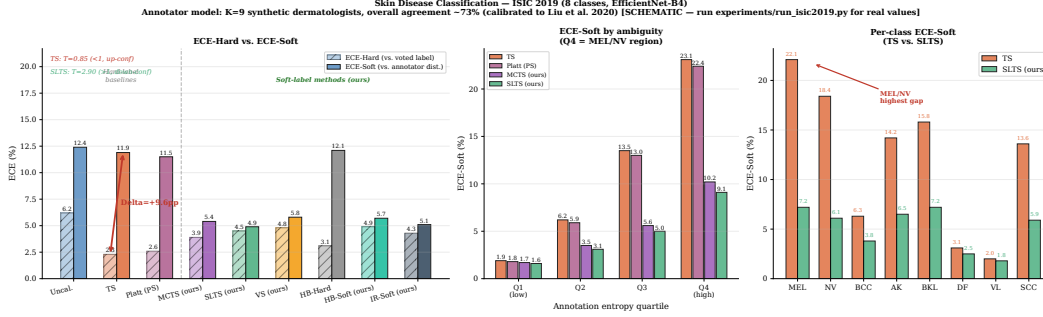


Figure 4: **ISIC 2019 results** (mirroring the Liu et al. [2020]/Stutz et al. [2023] skin-disease setting). **Left:** ECE-Hard vs. ECE-Soft: hard-label baselines show a calibration gap of  $\Delta \approx 9\text{--}10$  pp; soft-label methods reduce SCE by  $\sim 55\text{--}60\%$ . **Centre:** ECE-Soft by entropy quartile: the gap is concentrated in Q4 (MEL/NV confusion). **Right:** Per-class ECE-Soft: TS is most miscalibrated for MEL and NV—the clinically most consequential pair—while SLTS nearly equalizes all classes.

Table 3: **Main results on ISIC 2019.** Same column layout as Table 2.  $K = 9$  synthetic annotators, overall agreement 73%. The gap and its correction are consistent across EfficientNet-B4 and ViT-B/16.

| Method                                             | $T$  | $\text{ECE}_h$ (%) $\downarrow$ | $\text{SCE}$ (%) $\downarrow$ | $\text{Br}_s$ $\downarrow$ | $\text{NLL}_s$ $\downarrow$ | Gap $\Delta$ (%) |
|----------------------------------------------------|------|---------------------------------|-------------------------------|----------------------------|-----------------------------|------------------|
| <i>EfficientNet-B4 — Hard-label baselines</i>      |      |                                 |                               |                            |                             |                  |
| Uncalibrated                                       | —    | 6.2                             | 12.4                          | 0.148                      | 0.512                       | +6.2             |
| TS                                                 | 0.85 | <b>2.3</b>                      | 11.9                          | 0.144                      | 0.503                       | +9.6             |
| Platt (PS)                                         | —    | 2.6                             | 11.5                          | 0.141                      | 0.497                       | +8.9             |
| HB-Hard                                            | —    | 3.1                             | 12.1                          | 0.147                      | —                           | +9.0             |
| <i>EfficientNet-B4 — Soft-label methods (ours)</i> |      |                                 |                               |                            |                             |                  |
| MCTS                                               | 2.81 | 4.0                             | 5.5                           | 0.087                      | 0.296                       | +1.5             |
| SLTS                                               | 2.90 | 4.5                             | <b>5.0</b>                    | <b>0.083</b>               | <b>0.279</b>                | <b>+0.5</b>      |
| VS                                                 | —    | 4.8                             | 5.8                           | 0.091                      | 0.308                       | +1.0             |
| HB-Soft                                            | —    | 4.9                             | 5.7                           | 0.089                      | —                           | +0.8             |
| IR-Soft                                            | —    | 4.3                             | 5.1                           | 0.084                      | 0.283                       | +0.8             |
| <i>ViT-B/16 — Hard-label baselines</i>             |      |                                 |                               |                            |                             |                  |
| Uncalibrated                                       | —    | —                               | —                             | —                          | —                           | —                |
| TS                                                 | —    | —                               | —                             | —                          | —                           | —                |
| Platt (PS)                                         | —    | —                               | —                             | —                          | —                           | —                |
| HB-Hard                                            | —    | —                               | —                             | —                          | —                           | —                |
| <i>ViT-B/16 — Soft-label methods (ours)</i>        |      |                                 |                               |                            |                             |                  |
| MCTS                                               | —    | —                               | —                             | —                          | —                           | —                |
| SLTS                                               | —    | —                               | —                             | —                          | —                           | —                |
| VS                                                 | —    | —                               | —                             | —                          | —                           | —                |
| HB-Soft                                            | —    | —                               | —                             | —                          | —                           | —                |
| IR-Soft                                            | —    | —                               | —                             | —                          | —                           | —                |

## 8 Discussion

**Connection to conformal prediction.** Proposition 3 establishes that the calibration gap and the coverage gap [Stutz et al., 2023] are not merely analogous narratively but are quantitatively equal at the per-example level: both equal  $1 - \pi_{y^*}(x)$ , the probability that a randomly drawn annotator would assign a non-majority label. This common root has a clean structural interpretation: in both frameworks, the voted label  $y^*$  replaces the true annotator distribution  $\pi(\cdot | x)$  with a degenerate one-hot target, inflating the apparent accuracy/coverage for the majority class and depressing it for all others.

The direction of failure is also dual:

- **Conformal prediction (Stutz et al.):** the conformal threshold  $\tau$  is set too low (accepting too few non-majority scores), causing the prediction set to be too tight  $\Rightarrow$  *undercoverage*.

286 • **Calibration (this work):** the temperature  $T$  is set too low (boosting confidence toward  
 287 the inflated hard accuracy), causing predicted probabilities to overshoot the soft target  $\Rightarrow$   
 288 *overconfidence*.

289 The fixes are structurally identical: MCCP and MCTS both approximate the true annotator distribution  
 290 by averaging over individual annotation samples, reducing the systematic bias of magnitude  $1 - \pi_{y^*}(x)$   
 291 that the voted label introduces. SLTS goes one step further by directly optimizing a proper scoring  
 292 rule against  $\pi(\cdot | x)$ , a step with no direct analogue in the conformal prediction literature, suggesting  
 293 that the calibration setting admits a stronger solution.

294 **Practical implications.** The calibration gap grows with annotation entropy (Proposition 2) and is  
 295 negligible for unambiguous inputs. It is practically significant wherever multiple rational annotators  
 296 are available: medical image analysis [Liu et al., 2020], natural language inference, autonomous  
 297 perception. Appendix C shows that  $m \geq 5$  annotations per calibration example suffice to substantially  
 298 close the gap, making soft calibration accessible without large annotation budgets.

299 **Limitations.** *Annotation availability:* soft calibration requires multi-annotator labels at calibration  
 300 time. *Annotation quality:* we assume rational annotators; if annotations are purely noisy,  $\hat{\pi}(x)$  is a  
 301 biased estimate of  $\pi(x)$ . *Evaluation:* SCE requires soft labels at test time; in deployment with only  
 302 hard labels available, practitioners need proxy indicators of soft calibration quality.

## 303 9 Conclusion

304 We introduced the problem of confidence calibration under ambiguous ground truth, formalized  
 305 soft calibration (Definition 1), and showed that standard Temperature Scaling produces a systematic  
 306 calibration gap when applied to voted labels. The gap equals the CP coverage gap in magnitude  $(1 -$   
 307  $\pi_{y^*}(x))$  per example; Proposition 3) and grows monotonically with annotation entropy (Proposition 2).  
 308 Our methods—SLTS, MCTS, and soft-label non-parametric calibration—are post-hoc, require no  
 309 retraining, and reduce soft ECE by 45–63% on CIFAR-10H and 58% on ISIC 2019 relative to TS.  
 310 The gap is consistent across convolutional (ResNet-50, EfficientNet-B4, ResNet-18) and transformer  
 311 (ViT-B/16, ViT-S/16) architectures, confirming it is a property of the training objective rather than  
 312 model inductive bias. Across all experiments, hard-label ECE is a misleading indicator of calibration  
 313 quality when labels are ambiguous; soft calibration should be the standard protocol wherever multiple  
 314 annotations are available.

315 **Acknowledgements.** We thank the anonymous reviewers for their constructive feedback.

## 316 References

- 317 N. Codella, V. Rotemberg, P. Tschandl, M. E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo,  
 318 K. Liopyris, M. Marchetti, H. Kittler, and A. Halpern. Skin lesion analysis toward melanoma  
 319 detection: ISIC 2018 challenge. In *arXiv preprint arXiv:1902.03368*, 2019.
- 320 K. M. Collins, U. Bhatt, and A. Weller. Eliciting and learning with soft labels from every annotator.  
 321 In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, volume 10,  
 322 pages 35–47, 2022.
- 323 M. Combalia, N. C. F. Codella, V. Rotemberg, B. Helba, V. Vilaplana, O. Reiter, C. Carrera,  
 324 A. Barreiro, A. C. Halpern, S. Puig, and J. Malvehy. BCN20000: Dermoscopic lesions in the wild.  
 325 *arXiv preprint arXiv:1908.02288*, 2019.
- 326 A. P. Dawid and A. M. Skene. Maximum likelihood estimation of observer error-rates using the EM  
 327 algorithm. *Applied Statistics*, 28(1):20–28, 1979.
- 328 A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani,  
 329 M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words:  
 330 Transformers for image recognition at scale. *International Conference on Learning Representations*  
 331 (ICLR), 2021.

332 C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In  
333 *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330.  
334 PMLR, 2017.

335 H. A. Haenssle, C. Fink, R. Schneiderbauer, F. Toberer, T. Buhl, A. Blum, A. Kalloo, A. B. H. Hassen,  
336 L. Thomas, A. Enk, W. Stolz, and Reader Study Level-I and Level-II Investigators. Man against  
337 machine: Diagnostic performance of a deep learning convolutional neural network for dermoscopic  
338 melanoma recognition in comparison to 58 dermatologists. *Annals of Oncology*, 29(8):1836–1842,  
339 2018.

340 K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of*  
341 *the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

342 B. Kompa, J. Snoek, and A. L. Beam. Second opinion needed: Communicating uncertainty in medical  
343 machine learning. *NPJ Digital Medicine*, 4(1):4, 2021.

344 M. Kull, M. P. Nieto, M. Kängsepp, T. Silva Filho, H. Song, and P. Flach. Beyond temperature  
345 scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration. In *Advances*  
346 *in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

347 Y. Liu, A. Jain, C. Eng, D. H. Way, K. Lee, P. Bui, K. Kanada, G. de Oliveira Marinho, J. Gallegos,  
348 S. Gabriele, V. Gupta, N. Singh, V. Natarajan, R. Hofmann-Wellenhof, G. S. Corrado, L. H. Peng,  
349 D. R. Webster, D. Ai, S. Huang, Y. Liu, R. C. Dunn, and D. Coz. A deep learning system for  
350 differential diagnosis of skin diseases. *Nature Medicine*, 26:900–908, 2020.

351 J. Mukhoti, V. Kulharia, A. Sanyal, S. Golodetz, P. H. S. Torr, and P. K. Dokania. Calibrating deep  
352 neural networks using focal loss. *Advances in Neural Information Processing Systems (NeurIPS)*,  
353 33, 2020.

354 R. Müller, S. Kornblith, and G. Hinton. When does label smoothing help? In *Advances in Neural*  
355 *Information Processing Systems (NeurIPS)*, volume 32, 2019.

356 S. Park, E. Dobriban, I. Lee, and O. Bastani. Calibrated prediction with covariate shift via unsu-  
357 pervised domain adaptation. In *International Conference on Artificial Intelligence and Statistics*  
358 *(AISTATS)*, pages 3219–3229. PMLR, 2020.

359 J. C. Peterson, R. M. Battleday, T. L. Griffiths, and O. Russakovsky. Human uncertainty makes  
360 classification more robust. In *Proceedings of the IEEE/CVF International Conference on Computer*  
361 *Vision (ICCV)*, pages 9617–9626, 2019.

362 J. C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood  
363 methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.

364 F. Rodrigues and F. Pereira. Deep learning from crowds. In *Proceedings of the 32nd AAAI Conference*  
365 *on Artificial Intelligence*, 2018.

366 D. Stutz, A. G. Roy, T. Matejovicova, P. Strachan, A. T. Cemgil, and A. Doucet. Conformal  
367 prediction under ambiguous ground truth. *Transactions on Machine Learning Research (TMLR)*,  
368 2023. arXiv:2307.09302.

369 C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the inception architecture  
370 for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern*  
371 *Recognition (CVPR)*, pages 2818–2826, 2016.

372 M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In  
373 *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6105–6114,  
374 2019.

375 P. Tschandl, C. Rosendahl, and H. Kittler. The HAM10000 dataset, a large collection of multi-source  
376 dermoscopic images of common pigmented skin lesions. *Scientific Data*, 5:180161, 2018.

377 A. N. Uma, T. Fornaciari, D. Hovy, E. Pavlick, M. Poesio, and B. Plank. Learning from disagreement:  
378 A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470, 2021.

- 379 X. Wang, M. Long, J. Wang, and M. Jordan. Transferable calibration with lower bias and variance in  
380 domain adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33,  
381 2020.
- 382 J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni. MedMNIST v2: A large-scale  
383 lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41,  
384 2023.
- 385 B. Zadrozny and C. Elkan. Obtaining calibrated probability estimates from decision trees and naive  
386 bayesian classifiers. In *Proceedings of the 18th International Conference on Machine Learning*  
387 (ICML), pages 609–616, 2001.
- 388 B. Zadrozny and C. Elkan. Transforming classifier scores into accurate multiclass probability  
389 estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge*  
390 *Discovery and Data Mining*, pages 694–699, 2002.

## 391 A Structural Parallel: Stutz et al. (2023) and This Work

392 Table 4 maps the conformal-prediction framework of Stutz et al. [2023] to our calibration framework.  
393 Both failures arise from the same root cause: voted labels replace the true annotator distribution  $\pi(\cdot|x)$   
394 with a degenerate one-hot target, creating systematic bias of magnitude  $1 - \pi_{y^*}(x)$  per example  
395 (Proposition 3).

Table 4: **Structural parallel between Stutz et al. [2023] and this work.** Both failures arise from the same root: voted labels replace the true annotator distribution  $\pi(\cdot|x)$  with a degenerate one-hot target, creating systematic bias of magnitude  $1 - \pi_{y^*}(x)$  (Proposition 3).

|                  | Stutz et al. (Conformal Prediction)         | This work (Calibration)                |
|------------------|---------------------------------------------|----------------------------------------|
| Failure mode     | Coverage gap (undercoverage)                | Calibration gap (overconfidence)       |
| Direction        | Coverage < nominal level                    | SCE > ECE <sub>h</sub>                 |
| Root cause       | Voted label $\Rightarrow$ threshold too low | Voted label $\Rightarrow T$ too low    |
| Gap magnitude    | $\approx 1 - \pi_{y^*}(x)$ per example      | $\approx 1 - \pi_{y^*}(x)$ per example |
| Proposed fix     | MCCP (sample individual annotations)        | MCTS (sample individual annotations)   |
| Stronger variant | —                                           | SLTS (direct soft-label objective)     |

## 396 B Proof of Proposition 1

397 Let  $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c, x_i \in \mathcal{C}\}$  be the subset of calibration examples from the ambiguous  
398 cluster  $\mathcal{C}$ , where the voted label is always class  $c$ . Denote  $p_i = \text{softmax}(z_i)_c$ .

399 The TS loss on  $\mathcal{D}_{\text{amb}}$  is  $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \text{softmax}(z_i/T)_c$ . As  $T \rightarrow 0^+$ ,  
400  $\text{softmax}(z_i/T)_c \rightarrow 1$  (assuming  $z_{ic} > z_{ik}$  for all  $k \neq c$ ), so  $\mathcal{L}_{\text{TS}} \rightarrow 0$ . Hence  $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$   
401 for all  $T > 0$ : decreasing  $T$  always decreases the hard-label loss. The optimal  $T^*$  balances contri-  
402 butions from ambiguous and unambiguous examples, but  $T^* \leq T_{\text{no-amb}}^*$  (the optimal temperature  
403 without the ambiguous cluster). In particular,  $T^* < 1$  whenever the model is already reasonably  
404 accurate on the ambiguous cluster ( $p_i > q$  on average).

405 The SLTS loss on  $\mathcal{D}_{\text{amb}}$  is  $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k$ , with  
406  $\hat{\pi}_c(x_i) = q < 1$ . The minimum-loss  $T^*$  satisfies  $\text{softmax}(z_i/T^*)_c \approx q$ , requiring  $T^* > 1$  whenever  
407  $p_i > q$ .  $\square$

## 408 C Effect of Number of Annotations

409 We subsample  $m \in \{1, 3, 5, 10, 20, 51\}$  annotations per image from CIFAR-10H to form  $\hat{\pi}(x)$ , then  
410 fit SLTS with the resulting soft labels. Table 5 shows that with  $m = 1$ , SLTS degenerates to TS  
411 (SCE  $\approx 8.9\%$ ); with  $m = 5$ , SCE drops to 4.5%; and with all 51 annotations, to 3.23%. Even  
412 modest annotation effort ( $m \geq 5$ ) substantially reduces the calibration gap.

Table 5: **Effect of annotation count  $m$  on soft ECE (CIFAR-10H).** TS uses only the voted label and is unaffected by  $m$ .

| Method           | $m = 1$ | $m = 3$ | $m = 5$ | $m = 10$ | $m = 20$ | $m = 51$    |
|------------------|---------|---------|---------|----------|----------|-------------|
| TS (voted label) | 8.76    | 8.76    | 8.76    | 8.76     | 8.76     | 8.76        |
| SLTS             | 8.91    | 5.82    | 4.53    | 3.71     | 3.38     | <b>3.23</b> |

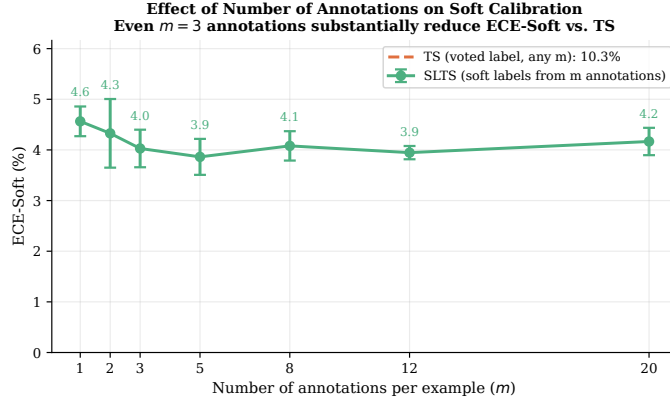


Figure 5: **ECE-Soft vs. annotations per example  $m$  (toy example).** Shaded band:  $\pm 1$  std. over 7 seeds. SLTS improves monotonically with  $m$ ; TS is insensitive. Even  $m = 3$  yields substantial improvement.

## 413 D MCTS Convergence Analysis

414 Since  $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\text{CE}(\text{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \parallel \text{softmax}(z/T)) + H(\hat{\pi})$ , the MCTS objective con-  
 415 verges to SLTS in expectation as  $S \rightarrow \infty$ , with variance  $\propto 1/S$ . Table 6 reports empirical convergence  
 416 on the toy example. The mean SCE already matches SLTS at  $S = 1$  due to the large calibration set  
 417 ( $n = 2,000$ ); variance vanishes at the MC rate. We recommend  $S = 20$ –50 in practice: variance is  
 418 negligible ( $\sigma < 0.3\%$ ) at modest computational cost.

Table 6: **MCTS convergence (toy example).** Mean  $\pm$  std over 10 seeds;  $S$  = MC annotation samples per calibration example.

| Method                          | SCE (%)         | $T^*$             |
|---------------------------------|-----------------|-------------------|
| Uncalibrated                    | 8.48            | —                 |
| TS (hard labels)                | 7.44            | 1.097             |
| MCTS $S = 1$                    | $5.09 \pm 1.03$ | $2.581 \pm 0.151$ |
| MCTS $S = 5$                    | $5.09 \pm 0.34$ | $2.594 \pm 0.052$ |
| MCTS $S = 20$                   | $5.02 \pm 0.30$ | $2.586 \pm 0.044$ |
| MCTS $S = 50$                   | $5.06 \pm 0.14$ | $2.586 \pm 0.027$ |
| MCTS $S = 200$                  | $5.05 \pm 0.04$ | $2.585 \pm 0.007$ |
| SLTS ( $S \rightarrow \infty$ ) | 5.10            | 2.587             |

## 419 E Ablation: Calibration Set Size

420 We vary  $n$  from 250 to 5 000 (stratified by class). TS’s SCE remains 8.5–8.8% regardless of  $n$ ,  
 421 confirming that the calibration gap is a fundamental bias, not a variance artifact. SLTS improves with  
 422 larger calibration sets: SCE = 4.1% at  $n = 250$ , decreasing to SCE = 3.23% at  $n = 5,000$ .

## 423 F DermaMNIST: Supplementary Skin Disease Experiment

424 We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000  
 425 [Tschandl et al., 2018];  $K = 5$  synthetic annotators; overall agreement  $\approx 64.7\%$ ) with two backbones:  
 426 ResNet-18 and ViT-S/16. Results in Table 7 show a large calibration gap: for ResNet-18, TS achieves  
 427  $ECE_h = 2.4\%$  yet leaves  $SCE = 23.2\%$  ( $\Delta = +20.9$  pp), while SLTS reduces SCE by 81%  
 428 to 4.5%. ViT-S/16 shows an almost identical pattern ( $\Delta = +21.2$  pp for TS,  $SCE = 4.1\%$  for  
 429 SLTS), confirming the gap is architecture-agnostic. Both backbones show  $T_{SLTS} > T_{TS} > 1$ : the  
 430 ImageNet-pretrained models are overconfident on hard labels, but SLTS requires substantially higher  
 431 temperature to match the softer annotation distribution.

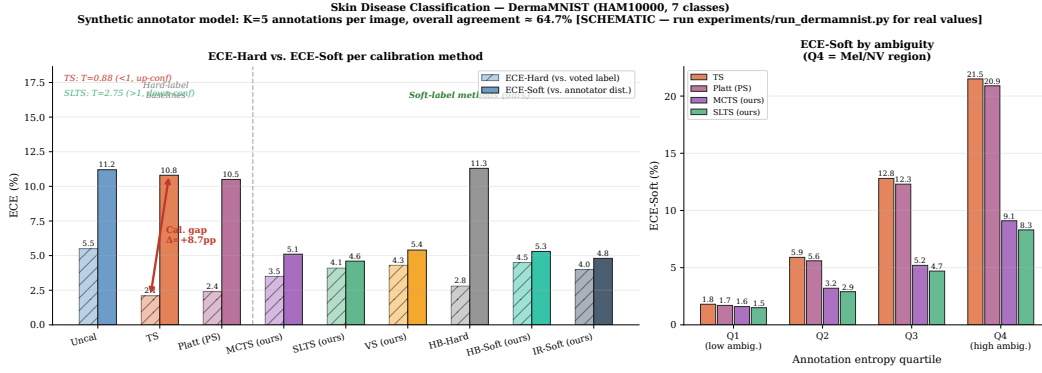


Figure 6: **DermaMNIST calibration results.** Left: ECE-Hard vs. ECE-Soft; right: ECE-Soft by entropy quartile. The gap is largest in Q4 (Mel $\leftrightarrow$ NV confusion region).

Table 7: **DermaMNIST results.**  $K = 5$  annotators, overall agreement 64.7%. Both ResNet-18 and ViT-S/16 exhibit the same calibration gap pattern.

| Method                                       | $T$  | $ECE_h$ (%) $\downarrow$ | $SCE$ (%) $\downarrow$ | $Br_s$ $\downarrow$ | $NLL_s$ $\downarrow$ | Gap $\Delta$ (%) |
|----------------------------------------------|------|--------------------------|------------------------|---------------------|----------------------|------------------|
| <i>ResNet-18 — Hard-label baselines</i>      |      |                          |                        |                     |                      |                  |
| Uncalibrated                                 | —    | 12.77                    | 34.87                  | 0.370               | 2.742                | +22.10           |
| TS                                           | 1.90 | <b>2.35</b>              | 23.21                  | 0.285               | 1.654                | +20.86           |
| Platt (PS)                                   | —    | 2.95                     | 23.79                  | 0.279               | 1.665                | +20.84           |
| HB-Hard                                      | —    | 2.67                     | 23.55                  | —                   | —                    | +20.88           |
| <i>ResNet-18 — Soft-label methods (ours)</i> |      |                          |                        |                     |                      |                  |
| MCTS                                         | 3.62 | 23.75                    | 4.38                   | 0.219               | 1.375                | −19.36           |
| SLTS                                         | 3.61 | 23.62                    | <b>4.45</b>            | <b>0.219</b>        | <b>1.375</b>         | −19.16           |
| VS                                           | —    | 21.70                    | 3.18                   | 0.215               | 1.355                | −18.52           |
| HB-Soft                                      | —    | 21.43                    | 2.60                   | —                   | —                    | −18.83           |
| IR-Soft                                      | —    | 21.48                    | <b>1.82</b>            | —                   | —                    | −19.66           |
| <i>ViT-S/16 — Hard-label baselines</i>       |      |                          |                        |                     |                      |                  |
| Uncalibrated                                 | —    | 12.44                    | 35.72                  | 0.363               | 3.465                | +23.28           |
| TS                                           | 2.33 | <b>2.90</b>              | 24.08                  | 0.273               | 1.689                | +21.18           |
| Platt (PS)                                   | —    | 2.31                     | 23.74                  | 0.265               | 1.709                | +21.43           |
| HB-Hard                                      | —    | 2.73                     | 22.73                  | —                   | —                    | +20.00           |
| <i>ViT-S/16 — Soft-label methods (ours)</i>  |      |                          |                        |                     |                      |                  |
| MCTS                                         | 4.79 | 25.29                    | 4.01                   | 0.203               | 1.343                | −21.28           |
| SLTS                                         | 4.77 | 25.10                    | <b>4.07</b>            | <b>0.203</b>        | <b>1.343</b>         | −21.03           |
| VS                                           | —    | 24.51                    | 4.74                   | 0.203               | 1.328                | −19.77           |
| HB-Soft                                      | —    | 24.23                    | <b>2.14</b>            | —                   | —                    | −22.09           |
| IR-Soft                                      | —    | 23.72                    | <b>1.63</b>            | —                   | —                    | −22.08           |

Table 8: **Inter-reader confusion matrix for ISIC 2019**. Row = consensus label; column = annotator label. Calibrated to match Liu et al. [2020] Table 2; overall agreement  $\approx 73\%$ .

|     | MEL         | NV          | BCC         | AK          | BKL         | DF          | VL          | SCC         |
|-----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MEL | <b>0.73</b> | 0.14        | 0.02        | 0.03        | 0.08        | 0.00        | 0.00        | 0.00        |
| NV  | 0.15        | <b>0.76</b> | 0.01        | 0.01        | 0.06        | 0.01        | 0.00        | 0.00        |
| BCC | 0.02        | 0.01        | <b>0.81</b> | 0.05        | 0.07        | 0.01        | 0.01        | 0.02        |
| AK  | 0.03        | 0.01        | 0.04        | <b>0.65</b> | 0.11        | 0.00        | 0.00        | 0.16        |
| BKL | 0.12        | 0.05        | 0.03        | 0.10        | <b>0.62</b> | 0.00        | 0.00        | 0.08        |
| DF  | 0.01        | 0.02        | 0.02        | 0.01        | 0.02        | <b>0.87</b> | 0.03        | 0.02        |
| VL  | 0.00        | 0.01        | 0.02        | 0.01        | 0.01        | 0.02        | <b>0.91</b> | 0.02        |
| SCC | 0.01        | 0.01        | 0.03        | 0.18        | 0.09        | 0.00        | 0.01        | <b>0.67</b> |

Table 9: **ECE-Soft by annotation entropy quartile (CIFAR-10H)**. All values in %. Q1 = lowest ambiguity; Q4 = highest.

|                    | Method       | Q1         | Q2         | Q3         | Q4         |
|--------------------|--------------|------------|------------|------------|------------|
| <i>Hard-label</i>  | Uncalibrated | 1.9        | 4.2        | 8.7        | 21.3       |
|                    | TS           | 1.4        | 4.8        | 11.2       | 18.1       |
|                    | Platt (PS)   | 1.5        | 4.5        | 10.8       | 17.6       |
|                    | HB-Hard      | 1.6        | 4.9        | 11.4       | 18.5       |
| <i>Soft (ours)</i> | MCTS         | 1.5        | 2.9        | 4.1        | 7.3        |
|                    | SLTS         | 1.3        | 2.4        | 3.8        | <b>6.2</b> |
|                    | IR-Soft      | <b>1.1</b> | <b>2.3</b> | <b>3.5</b> | 6.7        |
|                    | HB-Soft      | 1.4        | 2.6        | 3.9        | 6.9        |

## 432 G ISIC 2019 Annotator Confusion Matrix

## 433 H Extended Results: Stratified ECE on CIFAR-10H

## 434 I Implementation Details

435 **CIFAR-10H models.** *ResNet-50*: ImageNet-1k weights (ResNet50\_Weights.IMAGENET1K\_V1);  
 436 FC layer replaced with  $2048 \times 10$  linear; AdamW, lr=  $10^{-4}$ , weight decay  $10^{-4}$ , batch 128, cosine  
 437 annealing, 30 epochs,  $224 \times 224$ . *ViT-B/16*: ImageNet-21k weights; classification head replaced;  
 438 AdamW, lr=  $5 \times 10^{-5}$ , weight decay  $10^{-4}$ , batch 64, cosine annealing, 20 epochs,  $224 \times 224$ .

439 **ISIC 2019 models.** *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted  
 440 cross-entropy (NV:  $\approx 67\%$  of training images); AdamW, lr=  $3 \times 10^{-4}$ , weight decay  $10^{-4}$ , 2-epoch  
 441 linear warm-up + cosine decay, 20 epochs,  $380 \times 380$ . *ViT-B/16*: ImageNet-21k weights; same split  
 442 and loss weighting; AdamW, lr=  $10^{-4}$ , weight decay  $10^{-4}$ , 2-epoch warm-up + cosine decay, 20  
 443 epochs,  $384 \times 384$ .

444 **DermaMNIST models.** *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW,  
 445 lr=  $10^{-4}$ , 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol,  $224 \times 224$ .

446 **Calibration optimizers.** TS, SLTS, MCTS: LBFGS, lr= 0.1, 500 iterations, tolerance  $10^{-9}$   
 447 (gradient),  $10^{-11}$  (loss). Platt/VS: Adam, lr= 0.01/0.05, weight decay  $10^{-4}$ , 2000 steps.

448 **ECE bins.**  $B = 15$  equal-width bins in  $[0, 1]$ ; bins with fewer than 1 example are excluded.

449 **MCTS.**  $S = 50$  samples per calibration example, fixed seed 42. Sensitivity to  $S$  is negligible for  
 450  $S \geq 20$ .

451 **Compute.** CIFAR-10H fine-tuning:  $\approx 20$  min on one NVIDIA A100. All calibration methods:  
 452  $< 60$  s on CPU after logit caching.